Prediction

Compare 16s/ITS versus ME-NMR

```
phyXlong <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  transform_sample_counts(function(x) x / sum(x)) %>%
  filter_taxa(function(x) sum(x > 0)>10, TRUE) %>%
  psmelt() %>%
  dplyr::select(OTU:TSP, ta1:ta7) %>%
  full_join(MEconc_newlong %>%
              filter(!is.na(Participant)) %>%
              filter(`Blood origen`=='ME') ,
            by = c('Sample.ID' = 'Sample ID', 'Participant', 'Period')) %>%
  filter(!is.na(Abundance)) %>%
  filter(!is.na(logy)) %>%
  filter(!str_detect(met,'EREC')) %>%
  filter(!str_detect(met,'TSP')) %>%
  dplyr::select(OTU:ta7,met,category, type:logy)


tb_cor <- phyXlong %>%
  group_by(OTU,met,ta1,ta2,ta3,ta4,ta5,ta6,ta7) %>%
  do(cor.test(data = ., ~ logy + Abundance,method = 'spearman') %>% tidy)

tb_lmer <- phyXlong %>%
  # filter(met=='uridine') %>%
  # filter(OTU=='sp1') %>%
  mutate(logabu = log(Abundance+0.0001)) %>%
  group_by(OTU,met,ta1,ta2,ta3,ta4,ta5,ta6,ta7) %>%
  do(lmer(data = ., logy ~  logabu + factor(day) + (1|Participant) + (logabu|Participant)) %>

# tb_lmer %>% filter(str_detect(term,'log')) %>% arrange(p.value)
```

```
save(file = 'microbiome_vs_nmr_models.RData', list = c('tb_lmer', 'tb_cor','phyXlong'))
```

All versus all (16s vs NMR)

```
load(file = 'microbiome_vs_nmr_models.RData')

fmrt <- list(

  p.value = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  ),

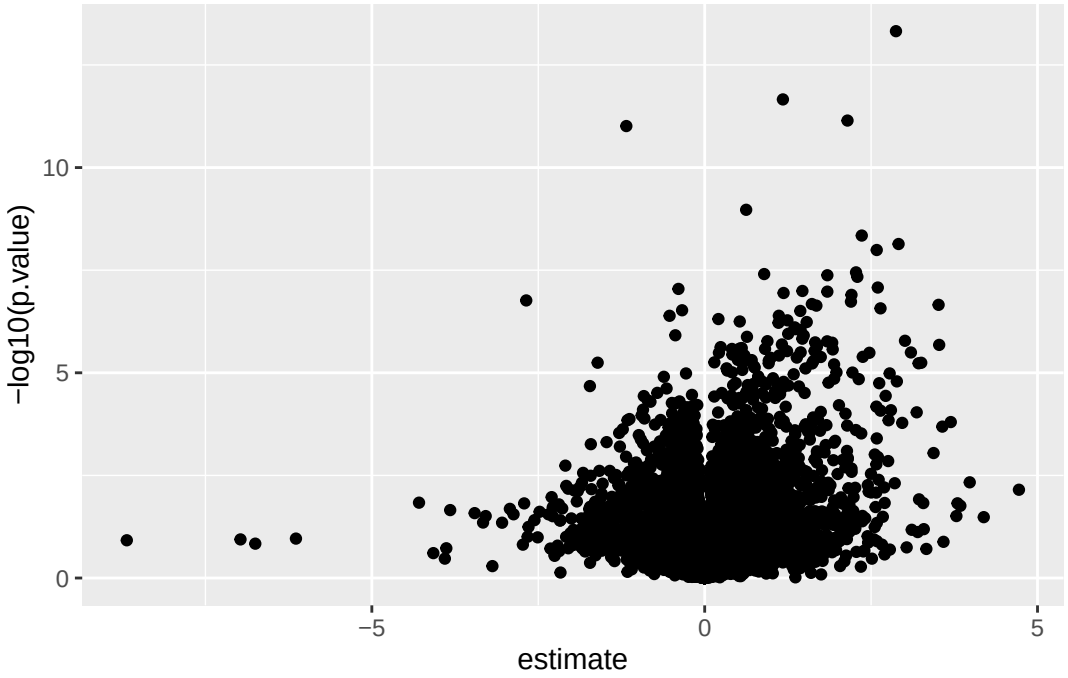
  estimate = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  ),

  std.error = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  )
)

tb_lmer2 <- tb_lmer %>%
  filter(term=='logabu') %>%
  ungroup %>%
  mutate(ta5 = ifelse(ta5 %in% c('F__','__'),'Other',ta5),
         ta5lmp = ta5 %>% fct_lump(n = 14),
         qval = p.adjust(p.value,'fdr'),
         pvcut = ifelse(p.value<0.05,'sign','ns'))

tb_lmer2 %>%
```

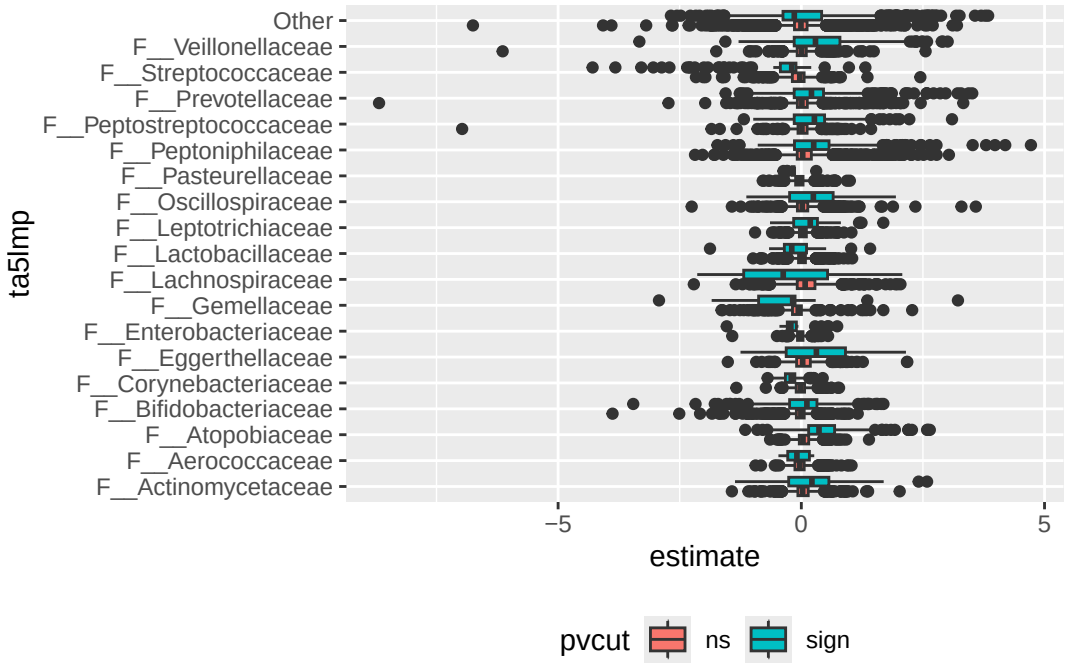
```
  ggplot(data =.,aes(estimate,-log10(p.value))) +
  geom_point()
```




```
tb_lmer2 %>%
  ungroup %>%
  dplyr::select(met,ta4,ta5,ta6,ta7,estimate,std.error, p.value) %>%
  reactable(columns = fmrt)
```

met	ta4	ta5	ta6	ta7	estimate	st
2-hydroxybutyric acid	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-3.05e-01	1
2-hydroxyisovaleric acid	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-9.39e-02	5
2-oxoisovaleric acid	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-2.53e-03	4
acetate	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-3.96e-01	1
alanine	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	8.66e-03	6
alpha-glucose	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	6.02e-02	8
aspartate	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	1.26e-02	7
bin-1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-2.09e-01	7
bin-1,2-propanediol	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	3.07e-02	6
bin-2	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-8.14e-02	4

```
tb_lmer2 %>% filter(str_detect(term,'log')) %>%
  ggplot(data =.,aes(ta5lmp, estimate,fill = pvcut)) +
  geom_boxplot() +
  theme(legend.position = 'bottom') +
  coord_flip()
```

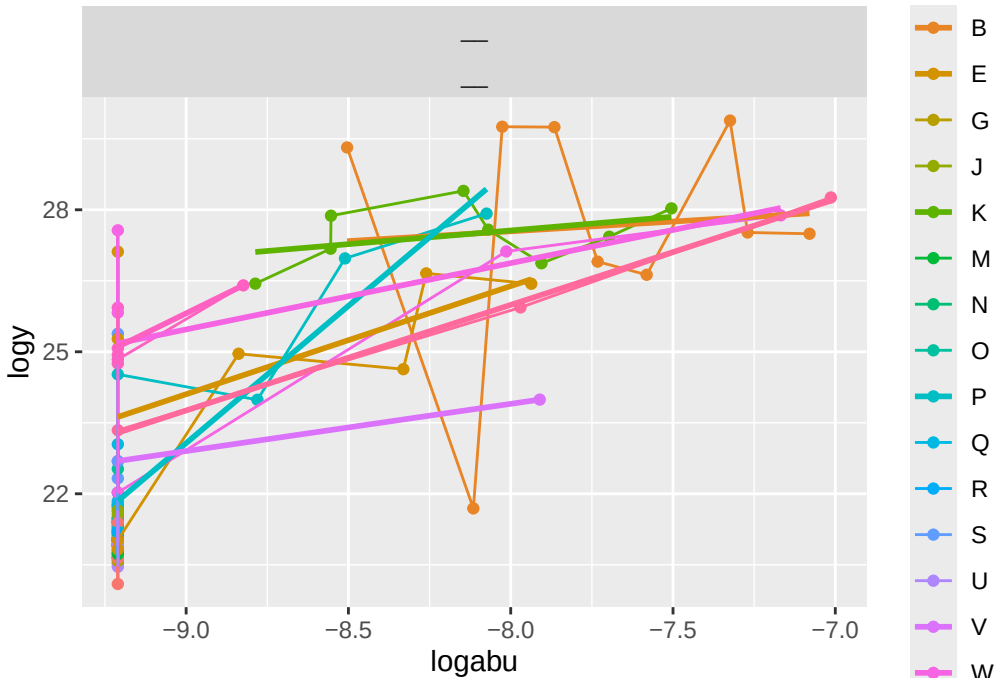



```
# tb_lmer %>% filter(str_detect(term,'log')) %>%
# group_by(ta5lmp) %>%
# dplyr::summarise(nneg = sum(estimate<0), npos = sum(estimate>0),
# pv = 1 - pbinom(max(npos,nneg)-1,nneg+npos,0.5))

gHM <- tb_lmer2 %>%
  group_by(OTU) %>%
  mutate(mq1 = min(qval)) %>%
  ungroup() %>%
  group_by(met) %>%
  mutate(mq2 = min(qval)) %>%
  filter(mq1<0.05 & mq2<0.05) %>%
  mutate(OTU2 = paste(ta4,ta5,ta6,ta7,OTU, sep = '') %>% factor()) %>%
  ggplot(data = ., aes(as.numeric(OTU2),substr(met,1,20),fill = estimate)) +
  geom_tile() +
  scale_fill_gradient2(mid = 'white', midpoint = 0)
```

```
ggsave(filename = 'Met_16s.pdf', plot = gHM, height = 20, width = 20)

phyXlong %>%
  # filter(met=='lactate') %>% filter(OTU=='sp145') %>%
  filter(met=='butyrate') %>% filter(OTU=='sp189') %>%
  mutate(logabu = log(Abundance+0.0001)) %>%
  ggplot(data = ., aes(logabu,logy, color = Participant, group = Participant)) +
  geom_point() + geom_line() + stat_smooth(method = lm, se = F) + facet_wrap(~ta6 + ta7)
```



Multivariate prediciton models using nested-cv

```
library(caret)
library(nestedcv)

make_loio_folds <- function(ids) {
  if (anyNA(ids)) {
    stop("ids must not contain missing values")
  }

  ids <- as.character(ids)
```

```
  n_ids <- length(unique(ids))

  if (n_ids < 3L) {
    stop("At least three unique IDs are required for nested leave-one-ID-out CV")
  }

  # Outer test folds: indices refer to rows in the complete dataset
  outer_folds <- split(
    seq_along(ids),
    factor(ids, levels = unique(ids))
  )

  # Inner test folds: indices refer to rows in each outer-training dataset
  inner_folds <- lapply(outer_folds, function(outer_test) {
    outer_train <- setdiff(seq_along(ids), outer_test)
    train_ids   <- ids[outer_train]

    split(
      seq_along(train_ids),
      factor(train_ids, levels = unique(train_ids))
    )
  })

  list(
    outer_folds = unname(outer_folds),
    inner_folds = lapply(inner_folds, unname)
  )
}
```

From 16s to NMR

```
phyXpp <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  subset_samples(!is.na(X2.hydroxyisovaleric.acid)) %>%
  transform_sample_counts(function(x) x / sum(x)) %>%
  filter_taxa(function(x) sum(x > 0)>10, TRUE)

X <- phyXpp@otu_table %>% data.frame()
X <- log(X + +0.0001)
Y <- phyXpp@sam_data %>% data.frame() %>% dplyr::select(-TSP,-ERECTIC)
```

```r
folds <- make_loio_folds(Y$Participant)

unmet <- unique(MEconc_newlong$met)
unmet <- unmet[!(unmet %in% c("ERECTIC","TSP"))]

for (i in unmet){
  print(i)
  yy <- MEconc_newlong %>% filter(met==i) %>% filter(!is.na(Participant)) %>% filter(`Blood c

  YY <- Y %>%
    left_join(yy, by = c('Participant','Period','Cycle.day' = 'Cycle day'))

  y <- YY$logy
  sdy <- sd(y)

  fit <- nestedcv::nestcv.train(
    x = X,
    y = y,
    method = "spls",                    # any caret model
    preProc = c("center", 'scale'),
    tuneGrid = expand.grid(K = 1:3,# K sets number of components
                           eta = seq(0,1,0.1), # eta sets sparsity
                           kappa = 0.5),
    outer_folds = folds$outer_folds,
    inner_folds = folds$inner_folds,
    finalCV = FALSE,
    cv.cores = 1
  )

  fit$sdy <- sdy
  fit$metabolite <- i

  flnm <- paste('./MLresults16s/metabolite_16smodels_nestedcv',
                i %>%
                  gsub('-','',x = .) %>%
                  gsub(' ','',x = .) %>%
                  gsub(',','',x = .),
                '_v2.RData', sep = '')
  save(file = flnm, list = c('fit','i'))
}
```

From ITS to NMR

```
phyITSpp <- phyITS %>%
  subset_samples(!is.na(Sample.ID)) %>%
  subset_samples(!is.na(X2.hydroxyisovaleric.acid)) %>%
  transform_sample_counts(function(x) as.numeric(x>0)) %>%
  filter_taxa(function(x) sum(x > 0)>9, TRUE)

X <- phyITSpp@otu_table %>% data.frame()
X <- log(X + +0.0001)
Y <- phyITSpp@sam_data %>% data.frame() %>% dplyr::select(-TSP,-ERECTIC)

folds <- make_loio_folds(Y$Participant)

unmet <- unique(MEconc_newlong$met)
unmet <- unmet[!(unmet %in% c("ERECTIC","TSP"))]
for (i in unmet){
  print(i)

  yy <- MEconc_newlong %>% filter(met==i) %>% filter(!is.na(Participant)) %>% filter(`Blood c
  YY <- Y %>%
    left_join(yy, by = c('Participant','Period','Cycle.day' = 'Cycle day'))

  y <- YY$logy
  sdy <- sd(y)

  fit <- nestedcv::nestcv.train(
    x = X,
    y = y,
    method = "spls",                  # any caret model
    preProc = c("center", 'scale'),
    tuneGrid = expand.grid(K = 1:2,# K sets number of components
                           eta = seq(0,1,0.1), # eta sets sparsity
                           kappa = 0.5),
    outer_folds = folds$outer_folds,
    inner_folds = folds$inner_folds,
    finalCV = FALSE,
    cv.cores = 1
  )

  fit$sdy <- sdy
  fit$metabolite <- i
```

```
  flnm <- paste('./MLresultsITS/metabolite_ITSmodels_nestedcv',
                i %>%
                  gsub('-','',x = .) %>%
                  gsub(' ','',x = .) %>%
                  gsub(',','',x = .),
                '_v2.RData', sep = '')
  save(file = flnm, list = c('fit','i'))
}
```

```
library(caret)
library(nestedcv)

flnm <- list.files('./MLresults16s/')
beta <- res16s <- data.frame()
for (fl in flnm){
  load(file = paste('./MLresults16s/',fl,sep = ''))
  k <- summary(fit)$final_param$K

  s <- summary(fit)$result %>% as.data.frame() %>%
    rownames_to_column('stat') %>%
    mutate(metabolite = i,method = '16s',sdy = fit$sdy, ncomp = k)
  s$x <- as.numeric(s$x)
  res16s <- bind_rows(res16s,s)


  beta <- beta %>% bind_rows(data.frame(breg = fit$final_fit$finalModel$betahat,
                                        metabolite = i,
                                        method = '16s',
                                        sdy = fit$sdy,
                                        ncomp = k
                                        ) %>%
                               rownames_to_column('OTU'))
}

flnm <- list.files('./MLresultsITS/')
resITS <- data.frame()
for (fl in flnm){
  load(file = paste('./MLresultsITS/',fl,sep = ''))
  k <- summary(fit)$final_param$K
  s <- summary(fit)$result %>% as.data.frame() %>%
    rownames_to_column('stat') %>%
    mutate(metabolite = i,method = 'ITS',sdy = fit$sdy, ncomp = k)
```

```
  s$x <- as.numeric(s$x)
  resITS <- bind_rows(resITS,s)

  beta <- beta %>% bind_rows(data.frame(breg = fit$final_fit$finalModel$betahat,
                                        metabolite = i,
                                        method = 'ITS',
                                        sdy = fit$sdy,
                                        ncomp = k
                                        ) %>%
                               rownames_to_column('OTU'))
}

load(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens

MLstat16s <- RMSE0 %>% left_join(res16s %>% filter(stat=='RMSE') %>% filter(!duplicated(metal
MLstatITS <- RMSE0 %>% left_join(resITS %>% filter(stat=='RMSE') %>% filter(!duplicated(metal

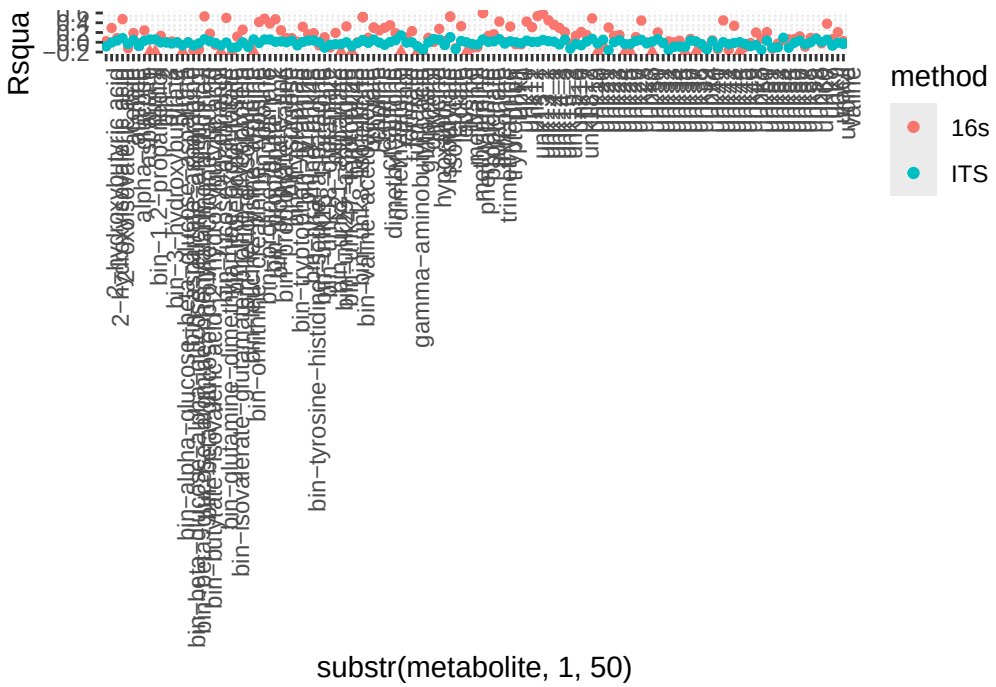
MLstat <- bind_rows(MLstat16s,MLstatITS) %>%
  mutate(stat = 'RMSE',
         ncomp = ifelse(is.na(x),0,ncomp),
         x = ifelse(is.na(x),rmse0,x)) %>%
  left_join(MEclass, by = c('metabolite' = 'Interval_name2'))

beta <- beta %>% left_join(MEclass, by = c('metabolite' = 'Interval_name2'))
save(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens
     list = c('MLstat','res16s','resITS', 'beta'))
```

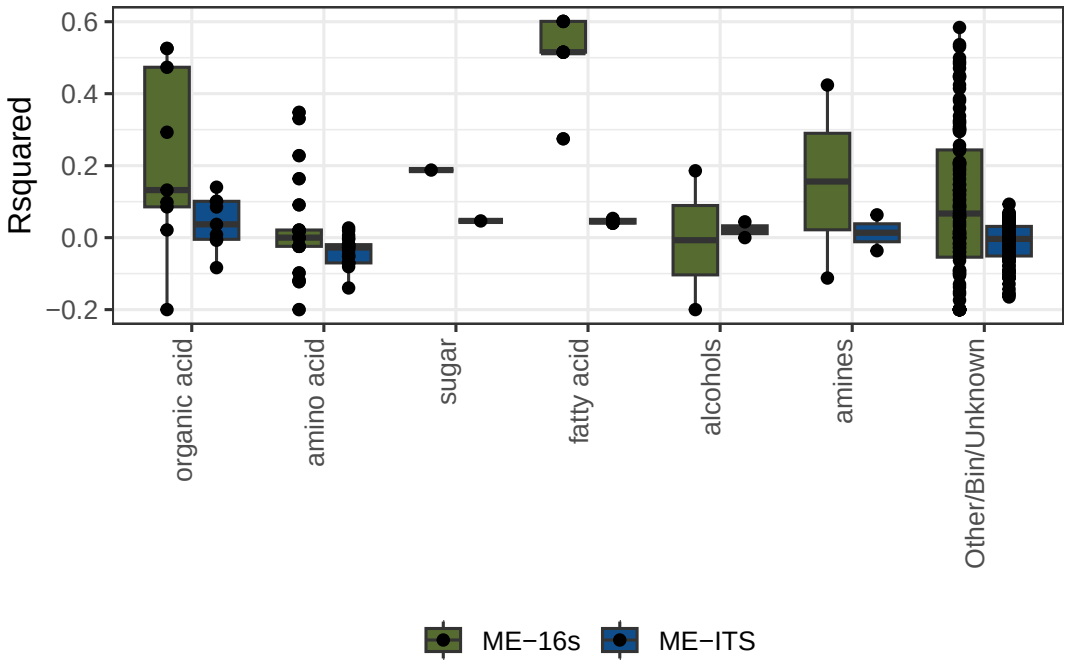
Per metabolite - CV predictive perormance.

```
load(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens

MLstat <- MLstat %>%
  mutate(
    Rsquared = 1 - (x^2 / rmse0^2),
    quality = ifelse(Rsquared < (-0.2),'overfitted','ok'),
    Rsquared = ifelse(Rsquared < (-0.2),-0.2,Rsquared))
MLstat %>%
  ggplot(data = ., aes(substr(metabolite,1,50),Rsquared, color = method, shape = quality)) +
  geom_point() + theme(axis.text.x = element_text(angle = 90, hjust = 1))
```




```
g_r2 <- MLstat %>%
  left_join(dflb, by = 'category') %>%
  filter(!is.na(lb)) %>%
  # group_by(lb) %>%
  # mutate(category2 = paste(lb, ' (#', n() / 2,')', sep = ''))  %>%
  # ungroup %>%
  mutate(lb = lb %>% factor(levels = dflb$lb[1:9])) %>%
  ggplot(data= ., aes(lb,Rsquared, fill = method)) +
  geom_boxplot() +
  # scale_y_reverse() +
  geom_point(position = position_dodge(width = 0.75)) +
  # geom_hline(data = data.frame(R2cut = c(R2sign_16s,R2sign_ITS), method = c('16s','ITS')),
  # aes(yintercept = R2cut, color = method)) +
  # geom_point(position = position_dodge(width = 0.8), aes(group = method), size = 1) +
  scale_fill_manual(values = colorcodes[1:2], labels = colorcodesnames[1:2]) +
  scale_color_manual(values = colorcodes[1:2], labels = colorcodesnames[1:2], guide = 'none')
  # scale_x_discrete(breaks = dflb$lb[1:9], values = dflb$lb[1:9]) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, vjust =  0),
        legend.position = 'bottom') + labs(fill = NULL) +
  xlab(NULL)
g_r2
```




```
MLstat %>%
  filter(!is.na(Rsquared)) %>%
  lm(rank(Rsquared) ~ method  + category ,data = .) %>% anova %>% tidy
```

```
# A tibble: 3 x 6
  term         df    sumsq  meansq statistic      p.value
  <chr>     <int>    <dbl>   <dbl>     <dbl>        <dbl>
1 method        1  187572. 187572.     28.6   0.000000175
2 category     14  403979.  28856.      4.41  0.000000369
3 Residuals   296 1938475.   6549.     NA    NA
```

```
MLstat %>%
  filter(!is.na(Rsquared)) %>%
  group_by(method) %>%
  summarise(R2m = median(Rsquared),
            q1 = quantile(Rsquared,0.25),
            q3 = quantile(Rsquared,0.75))
```

```
# A tibble: 2 x 4
  method      R2m      q1     q3
  <chr>     <dbl>   <dbl>  <dbl>
1 16s     0.0670  -0.0425 0.279
2 ITS    -0.00424 -0.0515 0.0337
```

Figure 5A in publication

```
library(gplots)

ggbeta <- beta %>%
  left_join(MLstat %>% dplyr::select(rmse0,x,Rsquared, metabolite,method,ncomp), by = c('meta
  left_join(dflb, by = 'category') %>%
  mutate(breg = breg/sdy) %>%
  filter(method == '16s') %>%
  filter(Rsquared>0.5) %>%
  left_join(tb_lmer2 %>% dplyr::select(OTU,met,estimate:pvcut), by = c('OTU','metabolite' =
  group_by(metabolite) %>%
  mutate(nn = sum(abs(breg)>0)) %>%
  filter(nn>0) %>%
  group_by(OTU) %>%
  mutate(nn2 = sum(abs(breg)>0)) %>%
  filter(nn2>0) %>%
  left_join(phyX@tax_table %>% data.frame() %>% rownames_to_column('OTU') %>%
              left_join(bsum16s, by = c('ta2','ta3','ta4','ta5')), by = 'OTU') %>%
  ungroup %>%
  mutate(taxall = paste(ta2,ta3,ta4,ta5, sep = '') %>% factor() %>% as.numeric(),
    OTU = OTU %>% fct_reorder(taxall)) %>%
  mutate(stars = gtools::stars.pval(qval))

lbs <- ggbeta %>%
  group_by(OTU) %>%
  dplyr::summarise(lb = ta5[1],
                   lb4 = ta4[1],
                   lb3 = ta3[1],
                   lb2 = ta2[1],
                   n()) %>%
  arrange(lb)

# 1) find lowest level of id, 2) remove F__
for (i in which(nchar(lbs$lb)<4)){
  id <- which(nchar(lbs[i,])>4)[1]
  if (is.na(id)) {
    lbs$lb[i] <- 'unassigned'
  }
  else {
  # print(as.character(lbs[i,id]))
  lbs$lb[i] <- as.character(lbs[i,id])
  }
```

```
}

lbs <- lbs %>%
  mutate(lb = lb %>%
           gsub('F__','',x = .) %>%
           gsub('__','-',x = .))


g5_all <- ggbeta %>%
  # remove bin from mtb label
  mutate(metabolite = gsub('bin.','',x = metabolite)) %>%
  # filter of only bins / unk
  # filter(nchar(metabolite)>6) %>%
  mutate(metabolite = gsub('X2.','2-',x = metabolite)) %>%
  # filter(!str_detect(metabolite,'unk')) %>%
  # filter(!str_detect(metabolite,'1_2')) %>%
  ggplot(data = ., aes(metabolite,OTU,fill = breg,label = stars)) +
  geom_tile() +
  geom_text() +
  scale_y_discrete(labels = lbs$lb,breaks =lbs$OTU ) +
  scale_fill_gradient2(mid = 'white',midpoint = 0) +
  theme_minimal() +
  theme(legend.position = 'top', axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0
  xlab(NULL) + ylab(NULL)

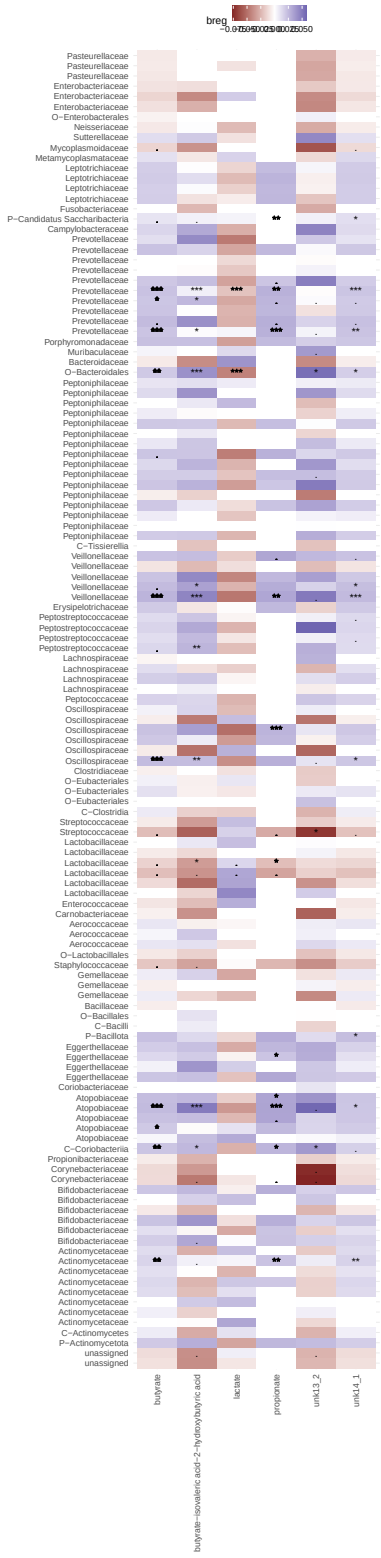

g5_sel <- ggbeta %>%
  # remove bin from mtb label
  mutate(metabolite = gsub('bin-','',x = metabolite)) %>%
  mutate(metabolite = gsub('1_2','',x = metabolite) %>% trimws()) %>%
  # filter of only bins / unk
  filter(nchar(metabolite)>6) %>%
  filter(!str_detect(metabolite,'unk')) %>%

  # filter(!str_detect(metabolite,'1_2')) %>%
  ggplot(data = ., aes(metabolite,OTU,fill = breg, label = stars)) +
  geom_tile() +
  scale_y_discrete(labels = lbs$lb,breaks =lbs$OTU ) +
  scale_fill_gradient2(mid = 'white',midpoint = 0) +
  geom_text() +
  # scale_color_manual(values = alpha(c('white','black'))) + #,c(0,1))) +
  theme_minimal() +
```

```
  theme(legend.position = 'top', axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0
  xlab(NULL) + ylab(NULL)
```

Coefficients - all

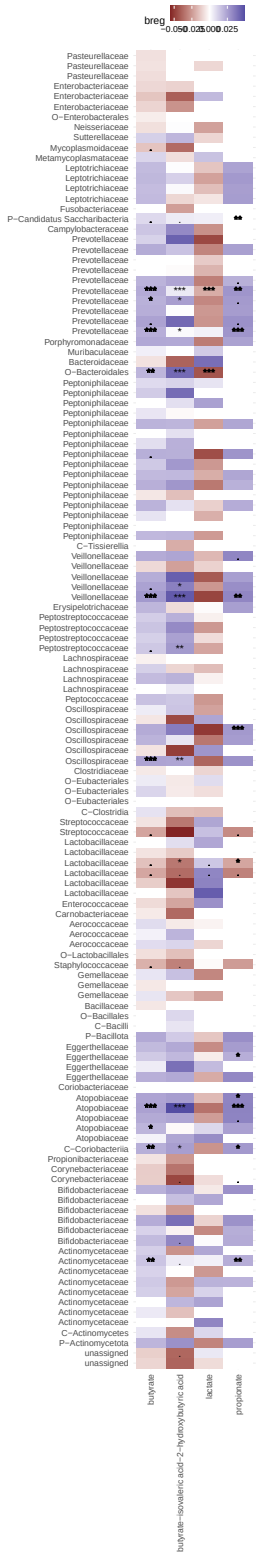
```
g5_all
```


breg
Pasteurellaceae
Pasteurellaceae
Pasteurellaceae
Enterobacteriaceae
Enterobacteriaceae
Enterobacteriaceae
O−Enterobacterales
Neisseriaceae
Sutterellaceae
Mycoplasmoidaceae
Metamycoplasmataceae
Leptotrichiaceae
Leptotrichiaceae
Leptotrichiaceae
Leptotrichiaceae
Fusobacteriaceae
P−Candidatus Saccharibacteria
Campylobacteraceae
Prevotellaceae
Prevotellaceae
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Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Porphyromonadaceae
Muribaculaceae
Bacteroidaceae
O−Bacteroidales
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
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Veillonellaceae
Erysipelotrichaceae
Peptostreptococcaceae
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Lachnospiraceae
Lachnospiraceae
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Oscillospiraceae
Oscillospiraceae
Clostridiaceae
O−Eubacteriales
O−Eubacteriales
O−Eubacteriales
C−Clostridia
Streptococcaceae
Streptococcaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Enterococcaceae
Carnobacteriaceae
Aerococcaceae
Aerococcaceae
Aerococcaceae
O−Lactobacillales
Staphylococcaceae
Gemellaceae
Gemellaceae
Gemellaceae
Bacillaceae
O−Bacillales
C−Bacilli
P−Bacillota
Eggerthellaceae
Eggerthellaceae
Eggerthellaceae
Eggerthellaceae
Coriobacteriaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
C−Coriobacteriia
Propionibacteriaceae
Corynebacteriaceae
Corynebacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
C−Actinomycetes
P−Actinomycetota
unassigned
unassigned
butyrate
butyrate−isovaleric acid−2−hydroxybutyric acid
lactate
propionate
unk13_2
unk14_1

Coefficients - selected

Figure 5B in publication

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g5_sel
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breg
−0.050
−0.025
0.000
0.025
Pasteurellaceae
Pasteurellaceae
Pasteurellaceae
Enterobacteriaceae
Enterobacteriaceae
Enterobacteriaceae
O−Enterobacterales
Neisseriaceae
Sutterellaceae
Mycoplasmoidaceae
Metamycoplasmataceae
Leptotrichiaceae
Leptotrichiaceae
Leptotrichiaceae
Leptotrichiaceae
Fusobacteriaceae
P−Candidatus Saccharibacteria
Campylobacteraceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Porphyromonadaceae
Muribaculaceae
Bacteroidaceae
O−Bacteroidales
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
C−Tissierellia
Veillonellaceae
Veillonellaceae
Veillonellaceae
Veillonellaceae
Erysipelotrichaceae
Peptostreptococcaceae
Peptostreptococcaceae
Peptostreptococcaceae
Peptostreptococcaceae
Lachnospiraceae
Lachnospiraceae
Lachnospiraceae
Lachnospiraceae
Peptococcaceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
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lactate
propionate